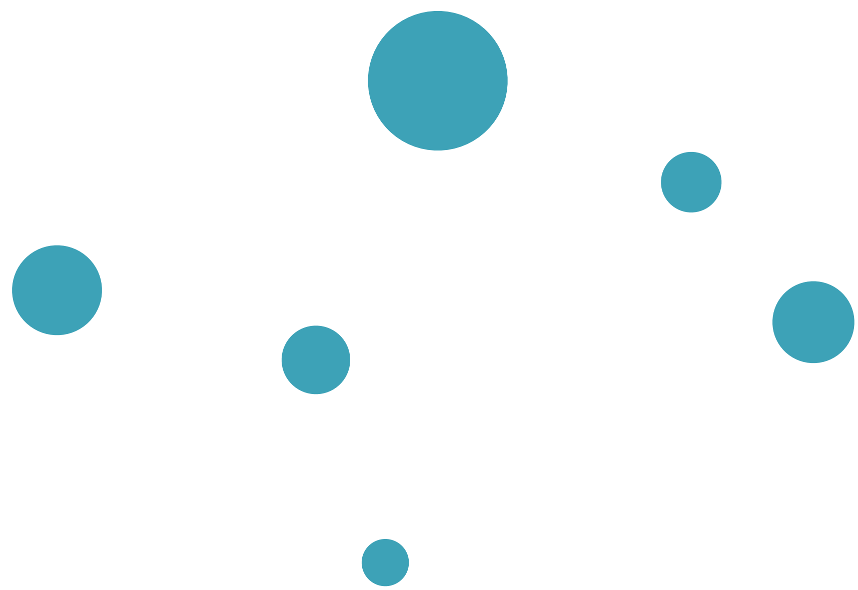
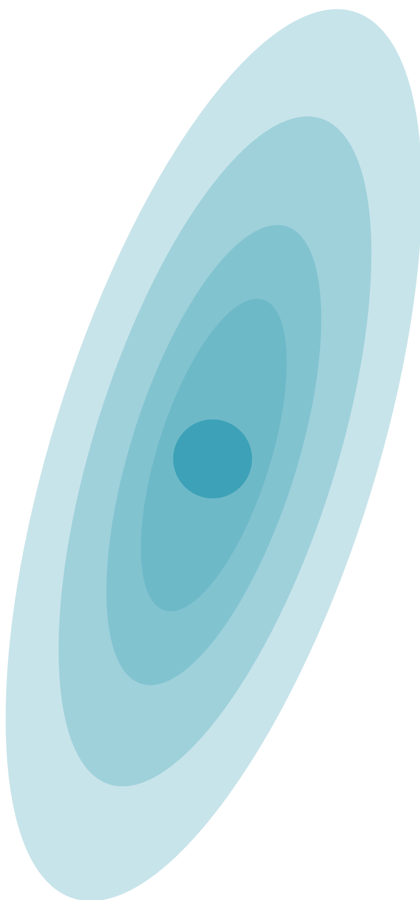
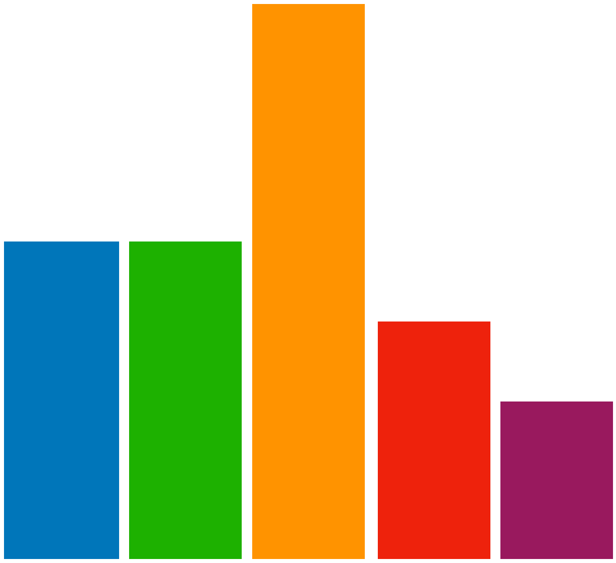


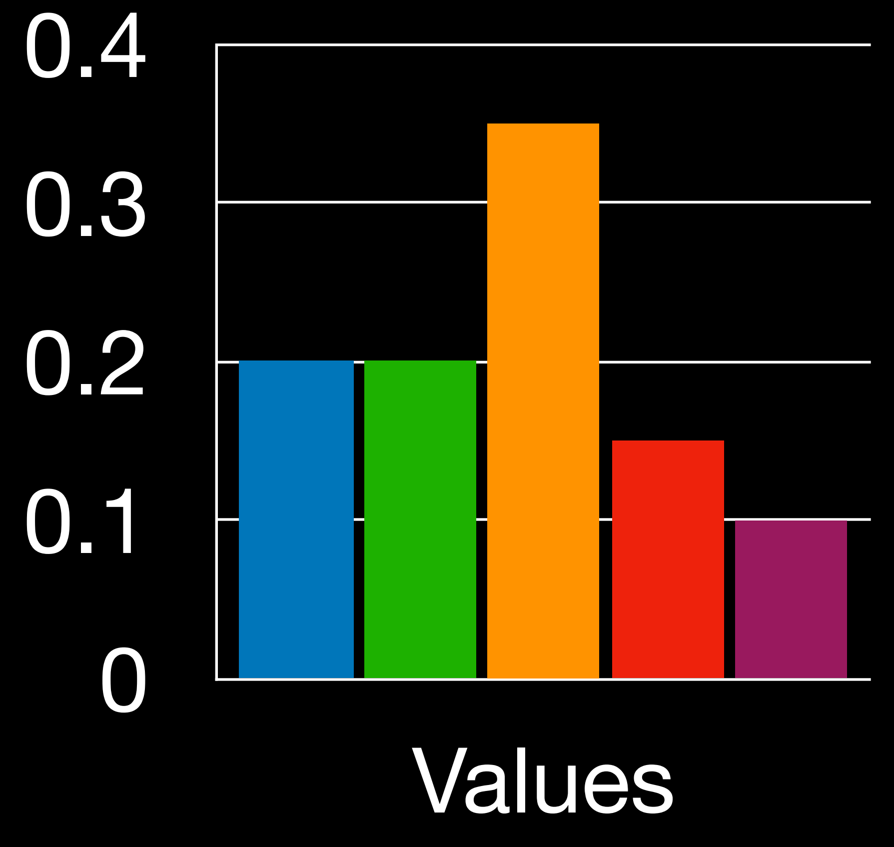
OT can deal with all three, agnostic to type, by working with **measures**





Three Types of Data

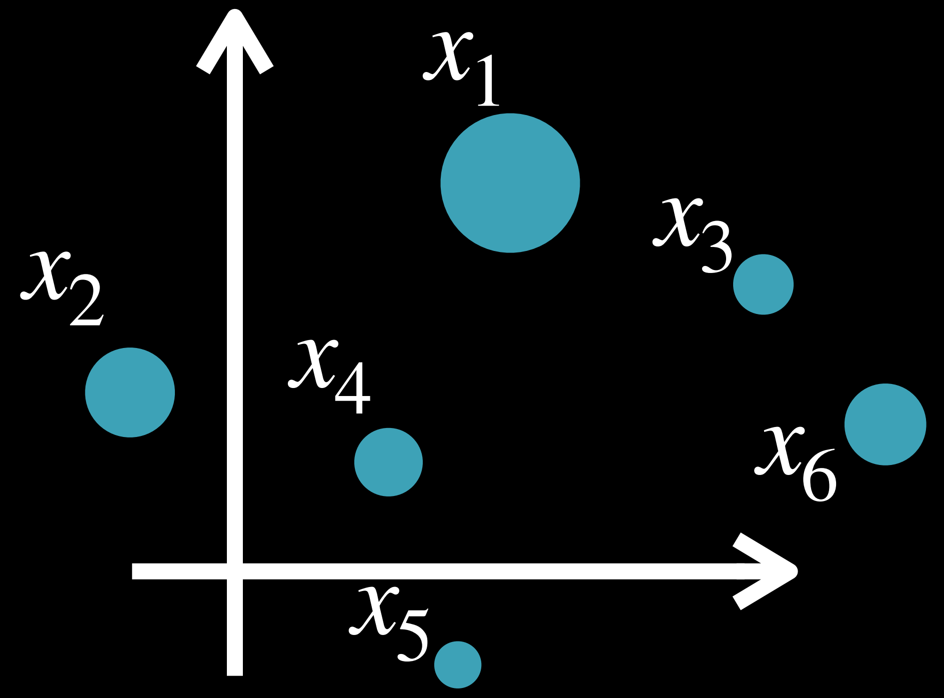
Probability Histograms



e.g. next-token probs, color profiles, pixel intensities

$$\mathbf{a} \in \mathbb{R}_+^n \ : \ \sum_{i=1}^{n} \mathbf{a}_i = 1$$

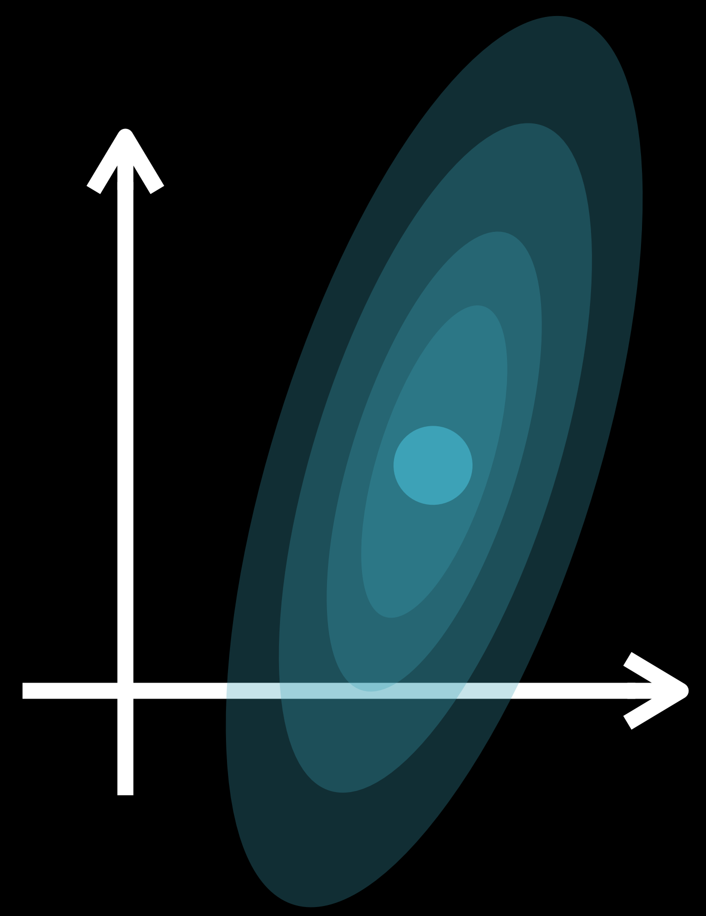
Discrete Distributions



e.g. word embeddings, NN activations, point clouds, 3D shapes

$$\alpha = \sum_{1}^{n} \mathbf{a}_i \delta_{x_i}; \sum_{i} a_i = 1$$

Continuous Distributions



e.g. generative models, noise distributions

$$\alpha = \int_{\mathcal{X}} f(x)\mathrm{d}x, \ \int f(x)\mathrm{d}x = 1$$

OT can deal with all three, agnostic to type, by working with **measures**

Two Ways to Discretize

Continuous Measure